Scenario-Based Quiz: Retrieval-Augmented Generation (RAG)

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Score: _____ / 20

Scenario 1: University Library Assistant

A university builds a **RAG chatbot** that answers questions from thousands of **digitized textbooks and research papers**. The system converts the documents into embeddings and stores them in a **vector database**.

1. When a student asks *"Explain supervised learning"*, what is the **first step** the RAG system performs?

A. Generate the answer directly
B. Retrieve relevant document chunks
C. Train a new model
D. Summarize the entire database

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2. Why does the system convert textbook content into **embeddings**?

A. To store images
B. To perform similarity search
C. To compress the database
D. To increase file size

Scenario 2: Legal Compliance Chatbot

A law firm creates a chatbot that answers questions based on **regulatory documents**. Each document is divided into **small chunks before embedding**.

3. Why is chunking necessary in this system?

A. To make documents visually attractive
B. To improve retrieval of relevant information
C. To increase storage cost
D. To convert text to speech

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4. If chunks are **too large**, what problem might occur?

A. Better accuracy always
B. Irrelevant information may be retrieved
C. Vector database will stop working
D. The LLM cannot generate answers

Scenario 3: Healthcare Knowledge Assistant

A hospital deploys a RAG system that retrieves medical guidelines and injects them into the LLM prompt before generating responses.

5. What is this step called?

A. Data preprocessing
B. Context injection
C. Data compression
D. Model training

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6. Why is context injection important?

A. It helps the LLM generate answers using relevant knowledge
B. It increases the dataset size
C. It replaces embeddings
D. It deletes irrelevant data

Scenario 4: Customer Support Chatbot

A telecom company builds a RAG chatbot trained on **FAQs, product manuals, and troubleshooting guides**.

Users complain that sometimes the chatbot gives **incorrect answers even though the correct document exists**.

7. Which issue is most likely causing this problem?

A. Poor retrieval quality
B. Too many GPUs
C. Large language model size
D. Internet speed

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8. Which evaluation metric would help diagnose the problem?

A. Retrieval relevance
B. Screen resolution
C. Disk usage
D. Network latency

Scenario 5: AI Research Assistant

An AI research lab uses a RAG system to summarize recent research papers. Engineers are experimenting with **different chunk sizes and retrieval strategies**.

9. What is the goal of testing different chunk sizes?

A. Reduce the number of documents
B. Optimize retrieval quality and context relevance
C. Remove embeddings
D. Replace the LLM

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10. Which step in the RAG pipeline happens **after retrieval**?

A. Tokenization of the dataset
B. Context injection into the prompt
C. Training the LLM
D. Deleting document chunks

✅ **Total Marks: 20**